Module 7a slides: Bioavailability and Bioequivalence (worked examples)

Module 7a - Bioavailability and Bioequivalence · Lecture slides, worked examples · 30 Sep 2026

Name _____ Date _____

Practice Problem

The bioavailability of an investigational drug was studied in 6 volunteers. Each volunteer received either a single oral tablet containing 250 mg of the drug or a single IV bolus injection containing 100 mg of the drug. The average AUC of the oral tablet was 101.5 (mcg/mL)hr and the average AUC of the IV bolus injection was 73.8 (mcg/mL)hr. What is the absolute bioavailability of the drug from the tablet?

Practice Problem (follow-up)

The absolute bioavailability of a drug from an oral tablet was found to be 0.55. A patient is receiving 100 mg of the drug by IV bolus. What oral dose of the tablet would provide the same extent of absorption as the 100 mg IV bolus dose?